```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.metrics import classification_report
```

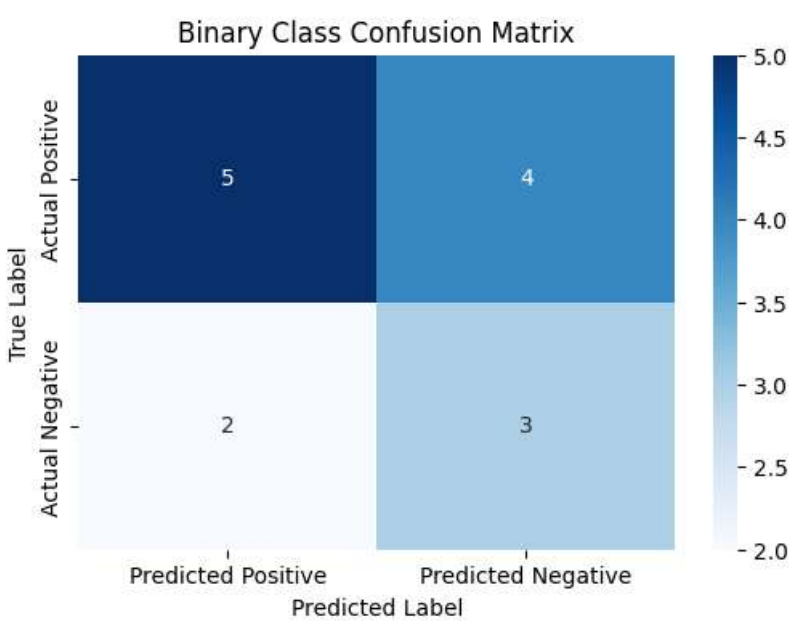
```
def binary_confusion_matrix(y_true, y_pred):
    TP = sum(np.logical_and(y_true, y_pred))
    TN = sum(np.logical_and(np.logical_not(y_true), np.logical_not(y_pred)))
    FP = sum(np.logical_and(np.logical_not(y_true), y_pred))
    FN = sum(np.logical_and(y_true, np.logical_not(y_pred)))
    return np.array([[TN, FP], [FN, TP]])

y_true = [1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0]
y_pred = [0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0]

conf_matrix = binary_confusion_matrix(y_true, y_pred)

# Adjust the confusion matrix to show TP in the top right corner
conf_matrix = np.flip(conf_matrix)
```

```
# Visualize confusion matrix
plt.figure(figsize=(6, 4))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues', xticklabels=['Predicted Positive', 'Predicted Negative'],
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Binary Class Confusion Matrix')
plt.show()
```



		Predicted	
		1	0
Actual	1	5 TP (11)	4 FN (10)
	0	2 FP (01)	3 TN (00)

Precision : $\frac{TP}{TP+FP} = \frac{5}{7} = 0.71$

Recall : $\frac{TP}{TP+FN} = \frac{5}{9} = 0.55$

F1_score : $\frac{2 \times P \times R}{(P+R)} = \frac{2 \times 0.71 \times 0.55}{0.71+0.55} = 0.61$

Actual = 1

F N

0 predicted

Actual relevant

classifier selected

falsely negative

```
# Convert lists to numpy arrays
y_true = np.array(y_true)
y_pred = np.array(y_pred)

# Generate classification report
target_names = ['Class 0', 'Class 1']
print(classification_report(y_true, y_pred, target_names=target_names))
```

```
              precision    recall  f1-score   support

     Class 0       0.43      0.60      0.50         5
     Class 1       0.71      0.56      0.63         9

    accuracy                           0.57        14
   macro avg       0.57      0.58      0.56        14
weighted avg       0.61      0.57      0.58        14
```